

**Automated Multi-Branch
Enterprise Network** with
Secure WAN and Monitoring

Project Team & Roles



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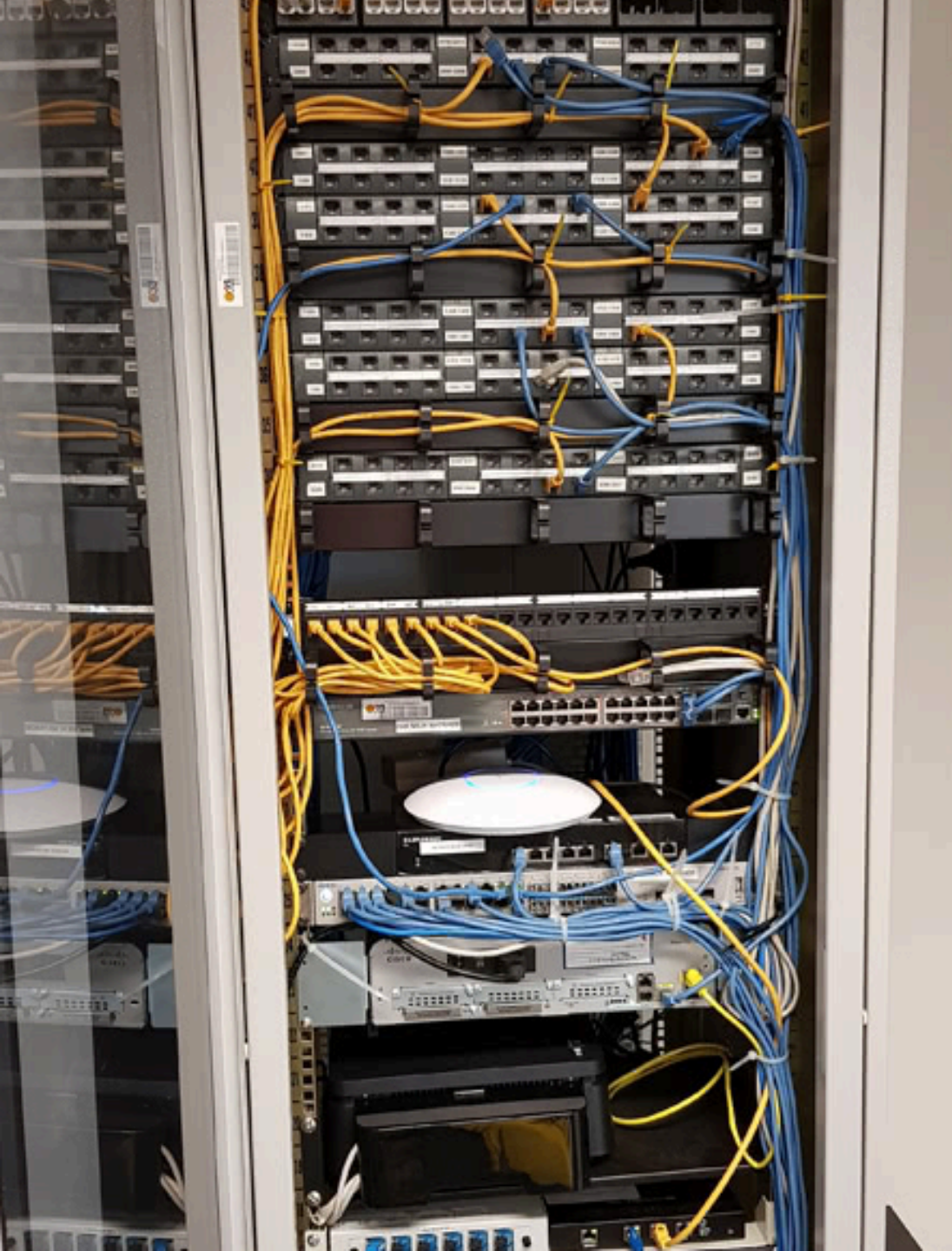
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Introduction

- Enterprises need secure and scalable networks
- Multi-branch connectivity must be reliable
- Manual management increases risk and complexity
- Automation and secure WAN are essential





Problem Statement

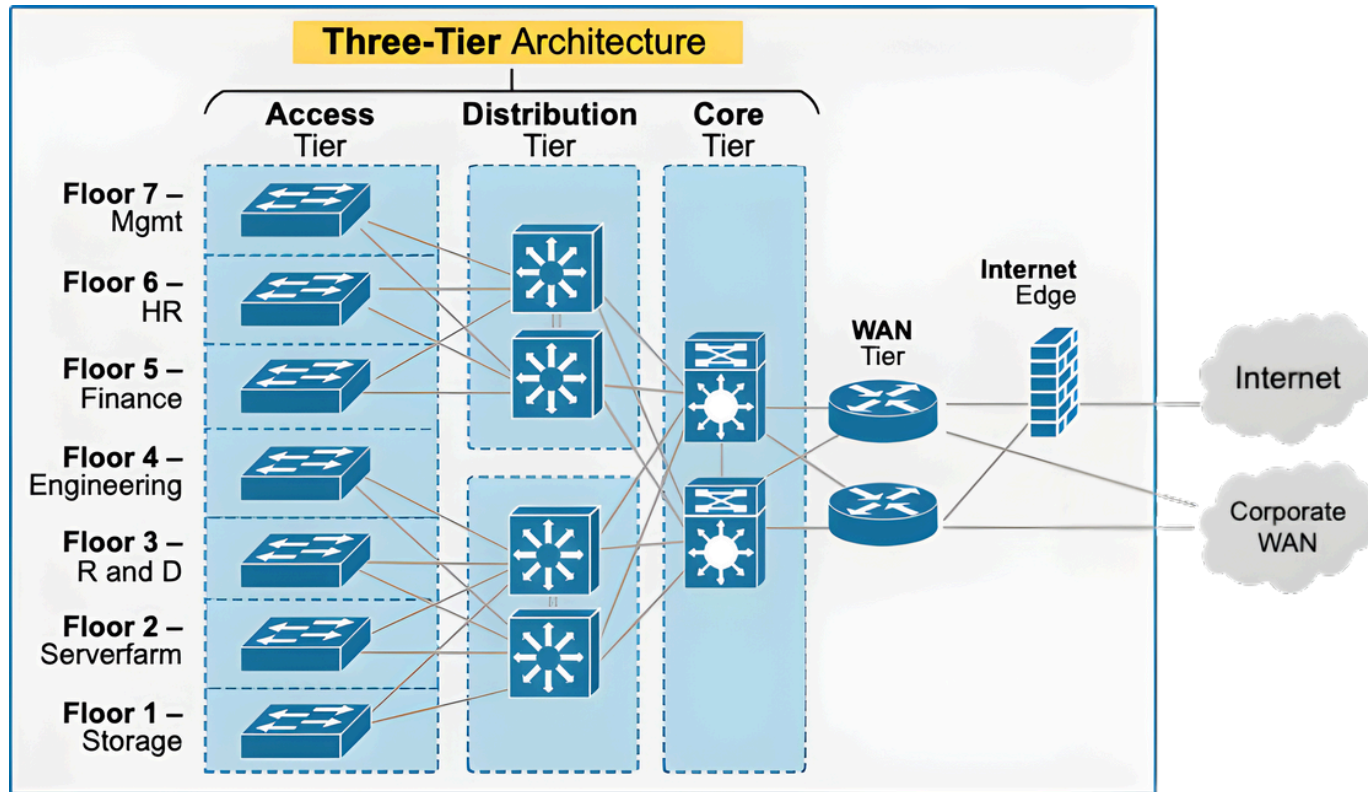
- **Network Congestion:** Inefficient inter-branch traffic flow
- **Security Risks:** Unsecured data transmission
- **Configuration Errors:** Manual configuration inconsistencies
- **Limited Visibility:** Lack of centralized monitoring
- **Scalability Issues:** Difficult branch expansion

Objectives

- **Design** a scalable and highly available multi-branch network.
- **Implement** secure WAN connectivity with centralized monitoring.
- **Automate** network configuration and management.



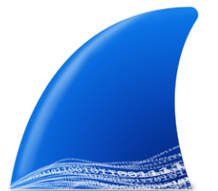
Proposed Network Design



- **Access layer** - Gives endpoints and users direct access to the network.
- **Distribution layer** - Provides an aggregation point for the access layer and acts as a services and control boundary between the access layer and the core layer.
- **Core layer** (also referred to as the backbone) - Provides connections between distribution layers for large environments.

Technologies & Tools

- PNetLab for enterprise network emulation
- Cisco IOS / IOS-XE for routing and switching
- IEEE 802.3, OSPF, VLAN, DHCP, NAT/PAT, RSTP, ICMP, EtherChannel, Port Security, and IPsec VPN for secure, scalable, and reliable connectivity.
- Python and Netmiko for network automation
- SNMP and Syslog for monitoring and logging
- MySQL and Linux Server for centralized data management



Conclusion

- The project demonstrates modern enterprise networking practices
- Automation improves efficiency and reliability
- Secure WAN ensures data protection
- The solution is suitable for real-world enterprise deployment



Questions?

We welcome your queries and insights to improve our project.